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Hypothesis testing for two independent samples

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Comparison of the means (expectations) of two populations

Case 1: the variances σ_1^2 and σ_2^2 are known

■ Assumptions:

- $X_1, X_2, \dots, X_n$ is a random sample chosen from a population-1 following a normal distribution $\mathcal{N}(\mu_1, \sigma_1^2)$;
- $Y_1, Y_2, \dots, Y_m$ is a random sample chosen from a population-2 following a normal distribution $\mathcal{N}(\mu_2, \sigma_2^2)$;
- The population-1 (presented by X) and the population-2 (presented by Y) are independent.
- In case 1: the variances σ_1^2 and σ_2^2 are known.

- At a given significant level α , the hypothesis testing for two independent populations are formed in one of the following 3 formulations:

$$(a) \begin{cases} H_0 : \mu_1 - \mu_2 = D_0 \\ H_1 : \mu_1 - \mu_2 \neq D_0 \end{cases} \quad \text{or} \quad (b) \begin{cases} H_0 : \mu_1 - \mu_2 = D_0 \\ H_1 : \mu_1 - \mu_2 < D_0 \end{cases} \quad \text{or} \quad (c) \begin{cases} H_0 : \mu_1 - \mu_2 = D_0 \\ H_1 : \mu_1 - \mu_2 > D_0 \end{cases}$$

Comparison of the means (expectations) of two populations

Case 1: the variances σ_1^2 and σ_2^2 are known

Hypothesis testing procedure

1. Formulate the null hypothesis H_0 and the alternative hypothesis H_1 .
2. Determine the significant level α .
3. From observed sample, calculate the statistic of test

$$Z_0 = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} \sim \mathcal{N}(0, 1). \quad (1)$$

Case 1: the variances σ_1^2 and σ_2^2 are known

4. Determine the critical region:

H_1	The critical region	p -value
$H_1 : \mu_1 - \mu_2 \neq D_0$	$ z_0 > z_{1-\frac{\alpha}{2}}$	$p\text{-value} = 2 \cdot [1 - \Phi(z_0)]$
$H_1 : \mu_1 - \mu_2 < D_0$	$z_0 < -z_{1-\alpha}$	$p\text{-value} = \Phi(z_0)$
$H_1 : \mu_1 - \mu_2 > D_0$	$z_0 > z_{1-\alpha}$	$p\text{-value} = 1 - \Phi(z_0)$

Table 1: The critical regions and p -values associating with H_1 .

5. Conclusion:

- If rejecting H_0 , we conclude that " H_1 is true" with confident level $(1 - \alpha)100\%$.
- Otherwise, we conclude that we do NOT reject H_0 with confident level $(1 - \alpha)100\%$.

Comparison of the means (expectations) of two populations

Case 1: the variances σ_1^2 and σ_2^2 are known - example

Example 3.1

A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are trialed: ten specimens are painted with formulation-1 which is the standard chemistry, and another ten specimens are painted with formulation-2 which has a new drying ingredient.

From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. The two sample mean of drying times are $\bar{x} = 121$ minutes and $\bar{y} = 112$ minutes, respectively.

What conclusions can the product developer draw about the effectiveness of the new ingredient, using $\alpha = 0,05$?

Answer:

1. Formulate the null hypothesis and the alternative hypothesis

$$\begin{cases} H_0 : \mu_1 - \mu_2 = 0, & \text{the new ingredient is ineffective,} \\ H_1 : \mu_1 > \mu_2, & \text{the new ingredient is effective.} \end{cases}$$

2. The significant level $\alpha = 5\% = 0.05$

Comparison of the means (expectations) of two populations

Case1: the variances σ_1^2 and σ_2^2 are known - example (cont.)

3. From observer sample, calculate the statistic of test:
with the sample means $\bar{x} = 121$, $\bar{y} = 112$ and $\sigma_1^2 = \sigma_2^2 = 8$, one has

$$z_0 = \frac{\bar{x} - \bar{y} - 0}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} = \frac{121 - 112}{\sqrt{\frac{8^2}{10} + \frac{8^2}{10}}} = 2.52.$$

4. Determine the critical region:
we will reject H_0 if $z_0 > z_{1-\alpha} = z_{0.95} = 1.65$.
5. Conclusion: as $z_0 = 2.52 > 1.65 = z_{1-\alpha}$, we reject H_0 .

We conclude that with confident level 95%, the new ingredient is effective to reduce the drying time.

- 5a. Method 2: using p -value: we have

$$p\text{-value} = 1 - \Phi(z_0) = 1 - \Phi(2.52) = 0.0059 < 0.05 = \alpha, \text{ so we reject } H_0.$$



Two independent populations:

- Comparison of the means (expectations) of two populations:
 - Case2: The variances σ_1^2 and σ_2^2 are unknown and large sample sizes

Comparison of the means (expectations) of two populations

Case 2: the variances σ_1^2 and σ_2^2 are unknown and large sample sizes

Assumptions:

- $X_1, X_2, \dots, X_n$ is a random sample chosen from a population-1, with the unknown mean (expectation) μ_1 and the unknown variance σ_1^2 .
- $Y_1, Y_2, \dots, Y_m$ is a random sample chosen from a population-2, with the unknown mean (expectation) μ_2 and the unknown variance σ_2^2 .
- The populations-1 (presented by X) and the population-2 (presented by Y) are independent.
- Large sample sizes: $n > 30$ and $m > 30$.

Comparison of the means (expectations) of two populations

Case2: the variances σ_1^2 and σ_2^2 are unknown and large sample sizes

- In this case, of large sample sizes, the variances of populations σ_1^2 and σ_2^2 are unknown, so we can use the sample variances S_1^2 and S_2^2 without making much change.
- When both $n > 30$ and $m > 30$, the statistic of test

$$Z_0 = \frac{\bar{X} - \bar{Y} - D_0}{\sqrt{\frac{S_1^2}{n} + \frac{S_2^2}{m}}} \quad (2)$$

can be approximately computed by the Gaussian distribution $\mathcal{N}(0, 1)$.

- The critical region (or p -value) in this case is calculated as in the case 1 of known variances (except for the replacement for σ_1 by S_1 and replacement for σ_2 by S_2).

Comparison of the means (expectations) of two populations

Case2: the variances σ_1^2 and σ_2^2 are unknown and large sample sizes - example

Example 3.2

Surveying on the height of students of the Faculty of Natural Science and Social Science: randomly select 50 students of the Faculty of Social Science found that the mean of height of 163cm and the standard deviation 5cm. Measuring the height of 50 students of the Faculty of Natural Science found that the mean of height of 166cm and the standard deviation 8cm.

At significant level 1%, draw a conclusion about the height of students in two faculties.

Answer:

Let $X(\text{cm})$ be the height of students of the Faculty of Social Science, so $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$, and let $Y(\text{cm})$ be the height of students of the Faculty of Natural Science, so $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$, moreover, $n = m = 50 > 30$.

1. Formulate the null hypothesis and the alternative hypothesis

$$\begin{cases} H_0 : \mu_1 - \mu_2 = 0, \\ H_1 : \mu_1 - \mu_2 \neq 0. \end{cases}$$

2. Significant level: $\alpha = 1\% = 0.01$.

Comparison of the means (expectations) of two populations

Case2: the variances σ_1^2 and σ_2^2 are unknown and large sample sizes - example (cont.)

Answer:(cont.)

3. From the observed sample, calculate the statistic of test:

We have $\bar{x} = 163$, $\bar{y} = 166$, $s_1 = 5$, $s_2 = 8$

and with $D_0 = \mu_1 - \mu_2 = 0$ (in the null hypothesis H_0), thus,

$$z_0 = \frac{163 - 166 - 0}{\sqrt{\frac{5^2}{50} + \frac{8^2}{50}}} = -2.248.$$

4. Determine the critical region:

we will reject H_0 if $|z_0| > z_{1-\frac{\alpha}{2}} = z_{0.995} = 2.58$.

5. Conclusion: as $|z_0| = |-2.248| = 2.248 < 2.58 = z_{1-\frac{\alpha}{2}}$, we do NOT reject H_0 .

We conclude with confident level $(1 - \alpha)100\% = 99\%$, that the height of students in the two faculties are similar.

- 5a. Method 2: using p -value, one has

$p\text{-value} = 2 \cdot [1 - \Phi(|z_0|)] = 2 \cdot [1 - \Phi(2.248)] = \dots < ? = 0.01 = \alpha$,

so we do NOT reject H_0 .

We conclude with confident level $(1 - \alpha)100\% = 99\%$, that the height of students in the two faculties are similar.

Two independent populations:

- Comparison of the means (expectations) of two populations:
 - Case 3: The variances σ_1^2 and σ_2^2 are unknown but with small sample sizes

Comparison of the means (expectations) of two populations

Case 3: The variances σ_1^2 and σ_2^2 are unknown but with small sample sizes

■ Assumptions:

- $X_1, X_2, \dots, X_n$ is a random sample chosen from a population-1 following a normal distribution with the unknown mean (expectation) μ_1 and the **unknown variance σ_1^2** .
- $Y_1, Y_2, \dots, Y_m$ is a random sample chosen from a population-2 following a normal distribution with the unknown mean (expectation) μ_2 and the **unknown variance σ_2^2** .
- **The population-1 (presented by X) and the population-2 (presented by Y) are independent.**
- **Small sample sizes:** $n \leq 30$ or $m \leq 30$.

■ We consider two circumstances:

1. In case of equal variances $\sigma_1^2 = \sigma_2^2$,
2. In case of different variances $\sigma_1^2 \neq \sigma_2^2$.

Comparison of the variances of two populations

■ Suppose that $X_1, X_2, \dots, X_n$ and $Y_1, Y_2, \dots, Y_m$ are respectively random samples chosen from two independent populations that has the mean and the variances $(\mu_1; \sigma_1^2)$ and $(\mu_2; \sigma_2^2)$, respectively. We want to test

$$\begin{cases} H_0 : \sigma_1^2 = \sigma_2^2, \\ H_1 : \sigma_1^2 \neq \sigma_2^2. \end{cases} \quad (3)$$

■ If S_1^2 is the sample variance of $(X_1, X_2, \dots, X_n)$, then

$$\frac{(n-1).S_1^2}{\sigma_1^2} \sim \chi^2(n-1), \quad (4)$$

similarly, one has

$$\frac{(m-1).S_2^2}{\sigma_2^2} \sim \chi^2(m-1). \quad (5)$$

Comparison of the variances of two populations

- Then, the statistic

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}, \quad (6)$$

has a Fisher distribution with $(n-1, m-1)$ degree of freedom.

- Consider a random variable $F \sim \mathcal{F}(u, v)$ having a probability density function $f(x)$, the upper quantile α of F denoted by $f_{\alpha, u, v}$ is defined as

$$\mathbb{P}(F > f_{\alpha, u, v}) = \int_{f_{\alpha, u, v}}^{+\infty} f(x) dx = \alpha. \quad (7)$$

- The upper quantile $(1-\alpha)$ of F is given by

$$f_{1-\alpha, v, u} = \frac{1}{f_{\alpha, u, v}}. \quad (8)$$

Comparison of the variances of two populations

Hypothesis testing procedure

1. Formulate the hypotheses $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_1 : \sigma_1^2 \neq \sigma_2^2$.
2. Determine the significant level α .
3. If H_0 is true, the statistic

$$F = \frac{S_1^2}{S_2^2}, \quad (9)$$

follows a Fisher distribution with $(n - 1, m - 1)$ degree of freedom.

4. Determine the critical region:

reject H_0 if $F > f_{\frac{\alpha}{2}, n-1, m-1}$ or $F < f_{1-\frac{\alpha}{2}, n-1, m-1}$.

5. Conclusion:

If rejecting H_0 , we conclude that H_1 is true with confident level $(1 - \alpha)100\%$.

Otherwise, we conclude that we do NOT reject H_0 .

Comparison of the means with small sample sizes, Case 3.1.: $\sigma_1^2 = \sigma_2^2 = \sigma^2$

- In case $\sigma_1^2 = \sigma_2^2 = \sigma^2$, we use a common estimate S_p^2 for both σ_1^2 and σ_2^2 . S_p^2 is called the pooled sample variance.

$$S_p^2 = \frac{(n-1).S_1^2 + (m-1).S_2^2}{n+m-2}. \quad (10)$$

- From observed sample, calculate the statistic of test

$$T_0 = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{S_p^2}{n} + \frac{S_p^2}{m}}} \quad (11)$$

follows a t -Student distribution with $(n+m-2)$ degree of freedom.

Comparison of the means with small sample sizes, Case 3.1.: $\sigma_1^2 = \sigma_2^2 = \sigma^2$

■ Set $df = n + m - 2$ (degree of freedom of t -Student random variable T_0), the critical region

H_1	The critical region	p -value
$H_1 : \mu_1 - \mu_2 \neq D_0$	$ t_0 > t_{1-\frac{\alpha}{2}}^{df}$	$p\text{-value} = 2.\mathbb{P}(T_{df} \geq t_0)$
$H_1 : \mu_1 - \mu_2 < D_0$	$t_0 < -t_{1-\alpha}^{df}$	$p\text{-giá tr} = \mathbb{P}(T_{df} \leq t_0)$
$H_1 : \mu_1 - \mu_2 > D_0$	$z_0 > t_{1-\alpha}^{df}$	$p\text{-value} = \mathbb{P}(T_{df} \geq t_0)$

Table 2: The critical regions and p -value corresponds to H_1 .

■ Conclusion: reject H_0 or do NOT reject H_0 .

Comparison of the means with small sample sizes, Case 3.2.: $\sigma_1^2 \neq \sigma_2^2$

- In case $\sigma_1^2 \neq \sigma_2^2$, the statistic of test is

$$T_0 = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n} + \frac{S_2^2}{m}}}. \quad (12)$$

- Then, T_0 follows a t -Student distribution with df degree of freedom which is given by

$$df = \frac{\left[\frac{s_1^2}{n} + \frac{s_2^2}{m} \right]^2}{\frac{\left(\frac{s_1^2}{n} \right)^2}{n-1} + \frac{\left(\frac{s_2^2}{m} \right)^2}{m-1}}. \quad (13)$$

- The critical region is similar to the case 3.1 of equal variances, except for the fact that the degree of freedom df is now compute by the formula (13).

Comparison of the means (expectations) of two populations

Case 3: the variances are unknown but with small sample sizes - example 1

Example 3.3

In a city, in an area A , select randomly 17 students for doing an EQ-test, found that the mean is 106 and the standard deviation is 10. Moreover, in an area B , the mean of EQ for a sample of 14 students is 109 with standard deviation 7. Suppose that the variances are equal to each other.

At a level of significance $\alpha = 0.02$, is there any difference in EQ of students in the areas A and B ?

Comparison of the means (expectations) of two populations

Case 3: the variances are unknown but with small sample sizes - example 1 (cont.)

Answer:

Let X be a r.v. presenting the EQ of students in area A , then $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$.

Let Y be a r.v. presenting the EQ of students in area B , then $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$.

With sample sizes $n = 17 < 30$ and $m = 14 < 30$.

According to the assumptions, we have $\sigma_1^2 = \sigma_2^2$.

1. Formulate the hypotheses: $H_0 : \mu_1 - \mu_2 = D_0 = 0$ vs $H_1 : \mu_1 - \mu_2 \neq 0$.

2. If H_0 is true, the statistic of test

$$T_0 = \frac{\bar{X} - \bar{Y} - D_0}{S_p \cdot \sqrt{\frac{1}{n} + \frac{1}{m}}} \sim T(n - 1 + m - 1) = T(29).$$

Comparison of the means (expectations) of two populations

Case 3: the variances are unknown but with small sample sizes - example 1 (cont.)

3. We have $\bar{x} = 106, \bar{y} = 109, s_1 = 10, s_2 = 7$ and $\mu_1 - \mu_2 = 0$, thus,

$$s_p = \sqrt{\frac{(n-1).s_1^2 + (m-1).s_2^2}{n-1+m-1}} = 8.733.$$

and

$$t_0 = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)}{s_p \cdot \sqrt{\frac{1}{n} + \frac{1}{m}}} = \frac{106 - 109 - 0}{8.733 \sqrt{\frac{1}{17} + \frac{1}{14}}} = -0.95.$$

4. With $\alpha = 0.02$, we will reject H_0 if $|t_0| > t_{1-\frac{\alpha}{2}}^{29} = t_{0.99}^{29} = 2.4620$, or using p -value

$p_{value} = 2 \cdot [1 - \mathbb{P}(T_{29} \leq | -0.95 |)]$, hence, $2 \cdot (1 - 0.90) < p_{value} < 2 \cdot (1 - 0.75)$.

5. We get $|t_0| = 0.95 < 2.4620 = t_{0.99}^{29}$, so $t_0 \notin W_\alpha$.
 (or one has $0.2 < p_{value} < 0.5$, so $p_{value} > 0.02$)

Thus, we do NOT reject H_0 , at significant level $\alpha = 0.02$.

Conclusion: with confident level $(1 - \alpha)100\% = 98\%$, the EQ of students in two areas are similar.

Comparison of the means (expectations) of two populations

Case 3: the variances are unknown but with small sample sizes - example 2

Example 3.4

Scientific researches shows that the higher Arsenic (unit: ppb) content in water, the more harmful it is to human health. The Arsenic content is examined in an area A in the city center and an area C near an airport.

In each area, the Arsenic content in water is measured randomly at 10 different locations. Collected data is given in the table below

city center	3	7	25	10	15	6	12	25	15	7
near the airport	48	44	40	38	33	21	20	12	1	18

With significant level 0.05, test whether there is a difference in the arsenic content in water in these two areas. Suppose that the variances are unequal.

Comparison of the means (expectations) of two populations

Case3: the variances are unknown but with small sample sizes - example 2 (cont.)

Answer:

Let X be a r.v. presenting the arsenic content at the area A (in city center),
 $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$.

Let Y be a r.v. presenting the arsenic content at the area B (near the airport),
 $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$.

The sample sizes are $n = m = 10 < 30$ and by assumptions $\sigma_1^2 \neq \sigma_2^2$.

1. Formulate the hypotheses: $H_0 : \mu_1 - \mu_2 = D_0 = 0$ vs $H_1 : \mu_1 - \mu_2 \neq 0$.
2. If H_0 is true, the statistic of test

$$T_0 = \frac{\bar{X} - \bar{Y} - D_0}{\sqrt{\frac{S_1^2}{n} + \frac{S_2^2}{m}}} \sim T(df).$$

Comparison of the means (expectations) of two populations

Case3: the variances are unknown but with small sample sizes - example 2 (cont.)

Answer:(cont.)

We have $\bar{x} = 12.5, \bar{y} = 27.5, s_1 = 7.634, s_2 = 13.35$ and $D_0 = 0$, so

$$df = \frac{\left[\frac{s_1^2}{n} + \frac{s_2^2}{m} \right]^2}{\left(\frac{s_1^2}{n} \right)^2 \frac{1}{n-1} + \left(\frac{s_2^2}{m} \right)^2 \frac{1}{m-1}} = \frac{\left(\frac{7.6346^2}{10} + \frac{15.35^2}{10} \right)^2}{\frac{\left(\frac{7.6346^2}{10} \right)^2}{9} + \frac{\left(\frac{15.35^2}{10} \right)^2}{9}} \approx 13.201.$$

Thus, $T_0 \sim T(13)$ the t -Student distribution with 13 degrees of freedom.

Moreover, we get

$$t_0 = \frac{\bar{x} - \bar{y} - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n} + \frac{s_2^2}{m}}} = -2.77.$$

4.Determine the critical region:

we will reject H_0 if $|t_0| > t_{1-\frac{\alpha}{2}}^{13} = t_{0.975}^{13} = 2.1604$.

5. Conclusion: since $|t_0| = 2.77 > 2.1604 = t_{1-\frac{\alpha}{2}}^{13}$, we reject H_0 ,

at significant level $\alpha = 0.05$.

Therefore, with confident level 95%, the arsenic content in water in these two areas are different from each other.

100

- 100

Comparison of the proportions of two populations

- Survey on individuals possessing a property $\mathcal{A}$ on two independent populations found that the corresponding proportions are p_1 and p_2 , respectively. From these two populations, choose two samples with sample sizes of n and m , respectively. Let X and Y be the number of individuals possessing the property $\mathcal{A}$ in the population-1 and in the population-2, respectively. Then, we have that $X \sim B(n, p_1)$ and $Y \sim B(m, p_2)$.
- Problem: compare the proportions p_1 and p_2 .
- The hypothesis testing problem is formed in one of the following 3 formulations:
(a) $\begin{cases} H_0 : p_1 - p_2 = D_0, \\ H_1 : p_1 - p_2 \neq D_0. \end{cases}$ (b) $\begin{cases} H_0 : p_1 - p_2 = D_0, \\ H_1 : p_1 - p_2 < D_0. \end{cases}$ (c) $\begin{cases} H_0 : p_1 - p_2 = D_0, \\ H_1 : p_1 - p_2 > D_0. \end{cases}$
- Assumptions:
 - The two populations under consideration are independent.
 - The sample sizes are reasonably large, satisfying $n.p_1 > 5, n.(1 - p_1) > 5$ and $m.p_2 > 5, m.(1 - p_2) > 5$.

Comparison of the proportions of two populations

Hypothesis testing procedure

- 1. Formulate the null hypothesis H_0 and the alternative hypothesis H_1 .
- 2. Determine the significant level α .
- 3. From observed sample, calculate the statistic of test

$$Z_0 = \frac{\hat{P}_1 - \hat{P}_2 - D_0}{\sqrt{\hat{P} \cdot (1 - \hat{P}) \cdot \left(\frac{1}{n} + \frac{1}{m}\right)}}, \tag{14}$$

with $\hat{P}_1 = \frac{X}{n}, \quad \hat{P}_2 = \frac{Y}{m} \quad \text{and} \quad \hat{P} = \frac{X + Y}{n + m}.$

If H_0 is true, then $Z_0 \sim \mathcal{N}(0, 1).$

Comparison of the proportions of two populations

4. Determine the critical region:

H_1	The critical region	p -value
$H_1 : p_1 - p_2 \neq D_0$	$ z_0 > z_{1-\frac{\alpha}{2}}$	$p\text{-value} = 2 \cdot [1 - \Phi(z_0)]$
$H_1 : p_1 - p_2 < D_0$	$z_0 < -z_{1-\alpha}$	$p\text{-value} = \Phi(z_0)$
$H_1 : p_1 - p_2 > D_0$	$z_0 > z_{1-\alpha}$	$p\text{-value} = 1 - \Phi(z_0)$

Table 3: The critical region and p -value correspond to H_1 .

5. Conclusion:

If rejecting H_0 , we conclude that H_1 is true with confident level $100 \times (1 - \alpha)\%$.
Otherwise, we conclude that we do NOT reject H_0 at significant level α .

Comparison of the proportions of two populations - example 1

Example 3.5

A pharmaceutical company needs to trial a new medicine that helps patients to reduce their chest pain. The company conducted trials on 400 patients, divided in two groups: group-1 consisting of 200 patients taking this new medicine while group-2 consisting of 200 patients taking a placebo.

Then, in group-1, there are 8 patients who have chest pain. On the other hand, in group-2, there are 25 patients who have chest pain.

At significant level $\alpha = 0,05$, draw a conclusion about the effectiveness of this new medicine.

Answer:

Let X and Y be r.v. presenting the number of patients who has chest pain in group-1 and in group-2, respectively. We have that $X \sim B(n, p_1)$ and $Y \sim B(n, p_2)$.

1. $H_0 : p_1 - p_2 = 0$ vs. $H_1 : p_1 - p_2 < 0$.
2. Determine the significant level: $\alpha = 0.05$.

Comparison of the proportions of two populations - example 1 (cont.)

Answer:(cont.)

3. If H_0 is true (i.e. $H_0 : p_1 - p_2 = D_0$), the statistic of test

$$Z_0 = \frac{\hat{P}_1 - \hat{P}_2 - D_0}{\sqrt{\hat{P} \cdot (1 - \hat{P}) \cdot \left(\frac{1}{n} + \frac{1}{m}\right)}} \sim \mathcal{N}(0, 1).$$

With sample proportions $\hat{p}_1 = \frac{8}{200} = 0.04$, $\hat{p}_2 = \frac{25}{200} = 0.125$ and $\hat{p} = \frac{8 + 25}{200 + 200} = 0.0825$, $D_0 = 0$ (as in $H_0 : p_1 - p_2 = D_0 = 0$), we get

$$z_0 = \frac{0.04 - 0.125 - 0}{\sqrt{0.0825 \times (1 - 0.0825) \times \left(\frac{1}{200} + \frac{1}{200}\right)}} = -3.0895.$$

4. The critical region: we will reject H_0 if $z_0 < -z_{1-\alpha} = -z_{0.95} = -1.65$.

5. Conclusion: since $z_0 = -3.095 < -1.65 = z_{1-\alpha}$, we will reject H_0 at significant level $\alpha = 0.05$.

We conclude that this new medicine has effect on reducing the possibility of getting chest pain, with confident level 95%.

Comparison of the proportions of two populations - example 2

Example 3.6

Randomly examining goods produced from two manufacturers and obtain the following data

- The manufacturer-1: there are 20 defective products out of a total of 1000 products.
- The manufacturer-2: there are 30 defective products out of at total of 900 products.

With significant level 0.02, can we conclude that the proportions of defective products from two manufacturers are similar?

Answer:

Let X and Y be r.v.s presenting the number of defective products from the manufacturer-1 and the manufacturer-2. Thus, $X \sim B(n, p_1)$ and $Y \sim B(m, p_2)$.

1. Formulate the hypotheses:

$$\begin{cases} H_0 : p_1 - p_2 = 0, & \text{proportions of defective products from 2 manufacturers} \\ & \text{are similar,} \\ H_1 : p_1 - p_2 \neq 0, & \text{proportions of defective products from 2 manufacturers} \\ & \text{are different.} \end{cases}$$

2. Significant level: $\alpha = 0.02$.

Comparison of the proportions of two populations - example 2 (cont.)

Answer:(cont.)3. If H_0 is true, the statistic of test:

$$Z_0 = \frac{\hat{P}_1 - \hat{P}_2 - D_0}{\sqrt{\hat{P} \cdot (1 - \hat{P}) \cdot \left(\frac{1}{n} + \frac{1}{m}\right)}} \sim \mathcal{N}(0, 1).$$

With $\hat{p}_1 = \frac{20}{1000} = 0.02$, $\hat{p}_2 = \frac{30}{900} \approx 0.0333$, $\hat{p} = \frac{20 + 30}{1000 + 900} \approx 0.0263$ and $D_0 = 0$, we have

$$z_0 = \frac{0.02 - 0.0333}{\sqrt{0.0263 \times (1 - 0.0263) \times \left(\frac{1}{1000} + \frac{1}{900}\right)}} = -1.7682.$$

4. Determine the critical region: this is two-sided hypothesis testing problem, so we will reject H_0 if $|z_0| > z_{1-\frac{\alpha}{2}} = 0.99 = 2.325$.

5. Conclusion: since $|z_0| = 1.7682 < 2.325 = z_{1-\frac{\alpha}{2}}$, we do NOT reject H_0 at a significant level 0.02.

Therefore, we can conclude that the proportions of defective products fro two manufacturers are similar, with confident level 95%.



